ATTNRES

Attention Residuals

Learned Softmax Retrieval across Network Depth | 2026

Replace uniform residual addition with input-dependent softmax attention over earlier layer outputs, allowing each layer to retrieve depth-wise information while controlling hidden-state growth.

residuals · depth attention · scaling

1 Problem

PRENORM DILUTION

- Standard residuals give every layer the same unit-weighted sum of all preceding outputs.
- With PreNorm, hidden magnitude grows with depth, diluting new layers and burying early information.
- Immediate-state recurrence offers no selective access to individual prior representations.

2 Full AttnRes

SOFTMAX OVER DEPTH

- Treat the embedding and each prior layer output as depth-wise key/value sources.
- Each layer owns one learned pseudo-query; normalized keys produce content-dependent weights.
- `h_l = sum_(il) v_i; sum alpha = 1`

3 Block AttnRes

PRACTICAL AT SCALE

- Partition L layers into N blocks, use ordinary residuals inside, and attend across block summaries.
- Memory and cross-stage communication fall from O(Ld) to O(Nd), retaining most gains.
- Cross-stage caching and two-phase online-softmax inference reduce traffic and amortize computation.

4 Learned behavior

SELECTIVE SKIP PATHS

- Weights stay locally diagonal but retain embeddings and sometimes reach far back—learned skip connections.
- Pre-attention layers use broader depth context; pre-MLP layers rely more sharply on recent states.
- Block aggregation bounds output magnitude and distributes gradients more uniformly.

5 Evidence

1.25x COMPUTE ADVANTAGE

- Block AttnRes matches a standard-residual baseline trained with 1.25x more compute.
- A 48B-total/3B-active Kimi Linear model trained on 1.4T tokens improves every evaluated task.
- GPQA-Diamond 36.9->44.4, Math 53.5->57.1 and HumanEval 59.1->62.2.

6 Cost & meaning

ATTENTION'S SECOND AXIS

- Training overhead is marginal and inference latency rises under 2%; about eight blocks recover most gains.
- AttnRes generalizes depth-wise linear residual mixing to softmax retrieval.
- Full access still costs O(Ld) memory; block granularity and very deep deployment remain open.